Rachit Kale

Software Developer | MERN Stack | GenAI Integration
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Education

K. J. Somaiya School of Engineering (KJSSE), Mumbai *2023 – 2027*
Final Year Project: Hybrid Anomaly Detection for Identifying Ambiguous UPI Transactions

Experience

Software Developer Intern — Synoris Information Systems Pvt Ltd. *Jun 2026 – Aug 2026*
Surat, India
MERN, Tailwind CSS, WebSockets, JWT auth, Postman, Gemini GenAI, GitHub

- **Owned end-to-end development** of a production-ready UPI transaction tracking system that captured and **logged payments in real time** via WebSockets, eliminating manual entry. **Chosen over alternative team proposals** for lower complexity and faster delivery
- Integrated Gemini API to auto-classify transactions into **9 spending categories**, powering an analytics module for automated reports, category-wise trends and predictive spend suggestions.
- Validated REST API integrations across frontend, backend and AI layers using Postman; shipped features through Git feature-branch workflow with iterative code reviews.

Software Development Intern — KJSSE (Institutional R&D) *Jan 2026 – Apr 2026*
Mumbai, India
MERN, Gemini API, GitHub, Vercel, Web Speech API, Monaco Editor, JWT auth

- Developed a **full-stack multimodal interview prep platform using MERN**, integrating the **Gemini API** to dynamically generate technical and behavioral questions based on user role, skills and uploaded resume
- Designed and contributed to RESTful APIs architecture for session management and automated feedback generation, **reducing feedback response time from 15s to 3-5s** through response optimization; collaborated on system architecture and testing for scalable deployment
- **Designed prompt structures and context injection** for Gemini API to generate personalized questions, iterating on prompts based on output quality and relevance.

Projects

Hybrid Anomaly Detection for Ambiguous UPI Transactions *Feb 2026 – Present*
React, Python, Apache Parquet, XGBoost, Random Forest, Isolation Forest, FastAPI

- Architected a hybrid fraud detection pipeline combining **supervised models (XGBoost, Random Forest)** with **unsupervised anomaly detection (Isolation Forest, LOF)** across 4 real world datasets, engineering a unified schema from heterogeneous transaction sources
- Designed an **explainable fusion scoring layer** combining dual model signals and uncertainty metrics to **flag ambiguous transactions** for review, avoiding a third opaque classifier
- Benchmarked models on a **9,00,000-transaction imbalanced dataset (6% fraud), achieving 0.95+ AUC-ROC** with class-weighted training and threshold tuning.

AI-Driven Train Induction Planning & Scheduling System *Sep 2025 – Nov 2025*
React.js, Node.js, PostgreSQL, AI Simulation, Data Visualization, DeepSeek API

- Designed and developed a interactive dashboard based decision support system for automated train induction planning for KMRL
- Developed an AI based simulation module to generate optimised train induction schedules using rule based constraints and multi objective logic
- Implemented explainable output generation with conflict alerts and "what-if" simulation capabilities
- Collaborated in a cross functional team environment to translate complex operational requirements into a scalable digital solution

Technical Skills

Languages: JavaScript, Typescript, SQL, HTML5, CSS3, Python
Backend & Database: Node.js, REST APIs, WebSockets, JWT Authentication, PostgreSQL, MongoDB
Frontend: React.js, Tailwind CSS
AI / LLM Integration: Gemini API, DeepSeek API, Prompt Engineering, LLM-based Classification
Tools: Git, GitHub

Certifications & Achievements

- Qualified for Round 2 — Smart India Hackathon 2025 (National Level); Developed AI-Driven Train Induction Planning & Scheduling system for Kochi Metro Rail Limited